1. What does HTML stand for in web technologies?

(A) HyperText Markup Language
(B) HighText Machine Language
(C) HyperTransfer Markup Level
(D) Hyperlink Text Manipulation Language

2. The standardized document type declaration for HTML5 at the top of a document is:

(A) <!DOCTYPE html>
(B) <!DOCTYPE HTML5>
(C) <!DOCTYPE html PUBLIC>
(D) <!DOCTYPE XHTML 1.0>

3. The root element that encloses all other HTML elements in a webpage is:

(A) <html>
(B) <root>
(C) <document>
(D) <body>

4. Metadata such as character encoding, page title, and stylesheet links are placed inside:

(A) <head>
(B) <body>
(C) <header>
(D) <meta>

5. Which HTML tag is used to specify the text displayed on the browser title bar or tab?

(A) <title>
(B) <heading>
(C) <meta>
(D) <caption>

6. Which meta tag declaration correctly specifies Unicode UTF-8 character encoding in HTML5?

(A) <meta charset="UTF-8">
(B) <meta encoding="UTF-8">
(C) <meta http-equiv="utf-8">
(D) <meta charset="ASCII">

7. Which element defines the visible content of an HTML document rendered in the browser window?

(A) <body>
(B) <main>
(C) <content>
(D) <section>

8. Which heading tag represents the highest hierarchical level of heading in HTML?

(A) <h1>
(B) <h6>
(C) <head>
(D) <header>

9. Which HTML tag creates a standard paragraph of text?

(A) <p>
(B) <para>
(C) <text>
(D) <pg>

10. Which empty (self-closing / void) element produces a forced line break in HTML?

(A)
(B) <lb>
(C) <break>
(D) <newline>

11. Which empty element creates a horizontal rule / thematic break between paragraphs?

(A) <hr>
(B) <line>
(C) <rule>
(D) <border>

12. The HTML anchor tag used to create hyperlinks to other web pages or internal bookmarks is:

(A) <a>
(B) <link>
(C) <href>
(D) <url>

13. Which attribute of the anchor tag <a> specifies the target URL destination?

(A) href
(B) src
(C) link
(D) target

14. To open a hyperlink in a new browser tab or window, what value should the target attribute be set to?

(A) target="_blank"
(B) target="_new"
(C) target="_parent"
(D) target="_self"

15. Which tag is used to embed an image into an HTML5 document?

(A) <img>
(B) <image>
(C) <pic>
(D) <src>

16. Which attribute of the <img> tag is essential for accessibility, providing descriptive fallback text?

(A) alt
(B) title
(C) caption
(D) desc

17. Which HTML tag creates an unordered (bulleted) list?

(A)
(B)
(C) <dl>
(D)

18. Which HTML tag creates an ordered (numbered) list?

(A)
(B)
(C) <nl>
(D)

19. Individual items inside ordered or unordered lists are defined using which tag?

(A)
(B) <item>
(C) <list>
(D)

20. A description list in HTML is defined using the tag:

(A) <dl>
(B)
(C) <ul-desc>
(D) <dd>

21. Inside a description list (<dl>), which tag defines the description term?

(A) <dt>
(B) <dd>
(C) <dl-item>
(D)

22. Inside a description list (<dl>), which tag defines the description definition/data?

(A) <dd>
(B) <dt>
(C) <def>
(D) <data>

23. Which HTML element is used to represent tabular data in rows and columns?

(A) <table>
(B) <grid>
(C) <tab>
(D) <tabular>

24. A table row inside an HTML table is defined using:

(A) <tr>
(B) <td>
(C) <th>
(D) <row>

25. Standard table data cells inside a row are created with:

(A) <td>
(B) <tr>
(C) <th>
(D) <cell>

26. Table header cells containing centered, bold text by default are created with:

(A) <th>
(B) <td>
(C) <thead-cell>
(D) <hcell>

27. Which HTML element provides a title or caption for an entire table?

(A) <caption>
(B) <title>
(C) <heading>
(D) <label>

28. Which attribute allows an HTML table cell to span horizontally across multiple columns?

(A) colspan
(B) rowspan
(C) span
(D) colwidth

29. Which attribute allows an HTML table cell to span vertically across multiple rows?

(A) rowspan
(B) colspan
(C) rowheight
(D) vspan

30. The semantic container used to group header rows in an HTML table is:

(A) <thead>
(B) <tbody>
(C) <tfoot>
(D) <th>

31. The container used to group the primary body rows in an HTML table is:

(A) <tbody>
(B) <thead>
(C) <tfoot>
(D) <tr>

32. The container used to group footer summary rows in an HTML table is:

(A) <tfoot>
(B) <thead>
(C) <tbody>
(D) <td>

33. Which HTML tag is used to create an interactive form for collecting user input?

(A) <form>
(B) <input-group>
(C) <fieldset>
(D) <submit>

34. Which form attribute specifies the HTTP method used to send form data (GET or POST)?

(A) method
(B) action
(C) enctype
(D) type

35. Which form attribute specifies the destination URL where submitted form data is processed?

(A) action
(B) method
(C) href
(D) target

36. Which input type creates a single-line plain text entry field?

(A) <input type="text">
(B) <input type="string">
(C) <input type="line">
(D) <input type="name">

37. Which input type masks user keystrokes with dots or asterisks for confidential entry?

(A) <input type="password">
(B) <input type="hidden">
(C) <input type="mask">
(D) <input type="secret">

38. Which input type creates a mutual exclusion group where only one option can be selected?

(A) <input type="radio">
(B) <input type="checkbox">
(C) <input type="select">
(D) <input type="toggle">

39. Which input type allows a user to select zero, one, or multiple independent options?

(A) <input type="checkbox">
(B) <input type="radio">
(C) <input type="multiple">
(D) <input type="button">

40. Which element creates a multi-line plain text input area for paragraphs or comments?

(A) <textarea>
(B) <input type="textbox">
(C) <input-multi>
(D) <p-input>

41. What is the primary conceptual difference between HTML4 layout tags (,) and HTML5 Semantic Elements (<article>, <section>, <nav>)?

(A) Semantic elements clearly convey their purpose and structure to browsers, search engines, and screen readers rather than generic division
(B) Semantic elements run faster on CPU
(C) Semantic elements are styled with CSS automatically without rules
(D) Semantic elements cannot contain images

42. Which HTML5 semantic element is designated for self-contained content that can be independently distributed or syndicated (e.g., blog post)?

(A) <article>
(B) <section>
(C) <aside>
(D) <main>

43. Which HTML5 element represents a standalone generic section of a document typically grouped with a heading?

(A) <section>
(B) <article>
(C)
(D) <part>

44. Which HTML5 semantic element encloses primary site navigation links?

(A) <nav>
(B) <menu>
(C) <header>
(D) <links>

45. Which HTML5 semantic element designates tangential, supplementary, or sidebar content related to surrounding content?

(A) <aside>
(B) <sidebar>
(C) <section>
(D) <extra>

46. Which HTML5 semantic element is used to encapsulate introductory content, site logo, or primary heading block?

(A) <header>
(B) <head>
(C) <top>
(D) <h1>

47. Which HTML5 semantic element encloses copyright info, author links, or document legal terms at the bottom of a page or section?

(A) <footer>
(B) <bottom>
(C) <end>
(D) <tfoot>

48. Which HTML5 element is reserved for the dominant, unique central content of the document, appearing only once per page?

(A) <main>
(B) <content>
(C)
(D) <body>

49. Which HTML5 element represents an illustration, diagram, code snippet, or photo referenced in the main flow?

(A) <figure>
(B) <pic>
(C) <illustration>
(D) <img>

50. Which element provides an explicit caption or legend for a <figure> element?

(A) <figcaption>
(B) <caption>
(C) <legend>
(D) <label>

51. Which native HTML5 element is used to embed audio files (e.g., MP3, WAV) directly into a web page?

(A) <audio>
(B) <sound>
(C) <media>
(D) <music>

52. Which native HTML5 element is used to embed video clips (e.g., MP4, WebM) without third-party plugins like Flash?

(A) <video>
(B) <movie>
(C) <media>
(D) <film>

53. Which boolean attribute must be added to <video> or <audio> tags to show standard play, pause, volume, and progress controls?

(A) controls
(B) autoplay
(C) show
(D) interface

54. Which HTML5 tag is used inside <audio> and <video> containers to specify multiple alternative media sources and formats?

(A) <source>
(B) <src>
(C) <track>
(D) <url>

55. Which HTML5 element specifies timed text tracks (such as subtitles, closed captions, and chapters) for video or audio media?

(A) <track>
(B) <subtitle>
(C) <caption-track>
(D) <vtt>

56. Which HTML5 form input type provides client-side validation to verify that an entered string matches an email syntax?

(A) <input type="email">
(B) <input type="mail">
(C) <input type="address">
(D) <input type="contact">

57. Which HTML5 form input type provides a numeric slider control bounded by min and max values?

(A) <input type="range">
(B) <input type="slider">
(C) <input type="scroll">
(D) <input type="level">

58. Which HTML5 form input type renders a native calendar or date picker widget in modern browsers?

(A) <input type="date">
(B) <input type="calendar">
(C) <input type="time">
(D) <input type="day">

59. Which HTML5 form input type allows selecting a color via a native color picker tool?

(A) <input type="color">
(B) <input type="palette">
(C) <input type="picker">
(D) <input type="rgb">

60. Which form element provides a predefined list of autocomplete suggestions linked to an <input> via the 'list' attribute?

(A) <datalist>
(B) <select>
(C) <options>
(D) <dropdown>

61. Which boolean attribute prevents form submission if a specific input field is left empty?

(A) required
(B) mandatory
(C) validate
(D) notnull

62. Which HTML5 attribute specifies a regular expression that the form input's value must match before submission?

(A) pattern
(B) regex
(C) validate
(D) match

63. Which attribute displays brief, light-gray instructional hint text inside an input field that disappears upon user typing?

(A) placeholder
(B) value
(C) title
(D) hint

64. Which form attribute in HTML5 specifies that an input field should automatically gain focus when the page loads?

(A) autofocus
(B) focus
(C) active
(D) first

65. What is the primary difference between HTTP GET and POST methods when submitting form data?

(A) GET appends data as URL query parameters in plain text; POST includes data securely inside the HTTP request body
(B) GET encrypts data; POST does not
(C) GET allows file uploads; POST does not
(D) GET handles larger payloads than POST

66. When creating a form that allows users to upload files (via <input type="file">), what enctype must be configured on the <form>?

(A) enctype="multipart/form-data"
(B) enctype="application/x-www-form-urlencoded"
(C) enctype="text/plain"
(D) enctype="binary/data"

67. Which HTML tag is used to draw scriptable 2D/3D pixel-based graphics via JavaScript?

(A) <canvas>
(B) <svg>
(C) <graphic>
(D) <draw>

68. How does Scalable Vector Graphics (<svg>) differ fundamentally from HTML5 <canvas>?

(A) SVG is XML-based vector format retained in the DOM; Canvas is a resolution-dependent pixel raster surface drawn imperatively
(B) SVG only draws 3D shapes
(C) Canvas is resolution independent
(D) SVG does not support CSS styling

69. Which HTML5 API provides persistent key-value client-side storage that survives browser restarts with no expiration date?

(A) localStorage
(B) sessionStorage
(C) IndexedDB
(D) Cookies

70. How does sessionStorage differ from localStorage in HTML5 Web Storage?

(A) sessionStorage data is cleared automatically when the browser tab or session window is closed
(B) sessionStorage can store 50 MB
(C) sessionStorage is shared across tabs
(D) sessionStorage encrypts data with RSA

71. Which HTML5 API allows a web application to prompt users for and obtain their geographical location coordinates?

(A) Geolocation API
(B) Google Maps API
(C) Positioning API
(D) GPS Tracker API

72. Which HTML5 element represents the completion progress of a background computational task or download?

(A) <progress>
(B) <meter>
(C) <bar>
(D) <status>

73. Which HTML5 element represents a scalar measurement within a known range, or a fractional value (e.g., disk storage usage)?

(A) <meter>
(B) <progress>
(C) <gauge>
(D) <level>

74. Which HTML5 interactive element creates a collapsible disclosure widget that can be opened or closed by clicking its summary?

(A) <details>
(B) <accordion>
(C) <expand>
(D) <collapse>

75. Which child element serves as the visible, clickable heading inside a <details> disclosure element?

(A) <summary>
(B) <caption>
(C) <header>
(D) <title>

76. What is the exact functional distinction between the tag and the <b> tag in modern semantic HTML5?

(A) implies strong semantic importance, gravity, or urgency for accessibility; <b> is purely visual bold styling without semantic weight
(B) <b> has higher SEO value
(C) only works on mobile
(D) They are 100% identical in all specs

77. Similarly, what is the functional difference between and <i> in HTML5?

(A) indicates semantic stress emphasis affecting speech inflection; <i> represents idiomatic alternate voice (e.g., technical terms, thoughts)
(B) <i> is deprecated in HTML5
(C) draws text in red
(D) makes text bold

78. Which HTML5 element is used to mark up computer program output, terminal logs, or calculator results?

(A) <samp>
(B) <code>
(C) <pre>
(D) <kbd>

79. Which HTML element represents user keyboard input, such as Ctrl+C or Enter key combinations?

(A) <kbd>
(B) <code>
(C) <samp>
(D) <key>

80. Which element represents machine-readable dates and times, enabling search engines and calendars to parse timestamps accurately?

(A) <time>
(B) <date>
(C) <datetime>
(D) <timestamp>

81. In the <time datetime="2026-10-26T14:30:00Z"> element, what format does the datetime attribute strictly adhere to?

(A) ISO 8601 standard date/time format
(B) UNIX epoch timestamp
(C) RFC 822 format
(D) Julian calendar format

82. What does the W3C Web Accessibility Initiative (WAI-ARIA) 'role' attribute achieve on HTML elements?

(A) It defines the structural role and widget purpose of an element to assistive technologies (screen readers) when semantic HTML is insufficient
(B) It overrides CSS styling classes
(C) It binds JavaScript click events
(D) It defines server backend routes

83. What is the purpose of ARIA live regions ('aria-live="polite"' or '"assertive"') on dynamic web elements?

(A) Informing screen readers to announce real-time asynchronous updates to users without interrupting current speech (polite) or immediately (assertive)
(B) Caching audio buffers
(C) Running live video feeds
(D) Encrypting dynamic forms

84. Why should the <label> element always be bound to form controls via the 'for' attribute matching the input's 'id'?

(A) It maximizes hit targets (clicking label focuses input) and ensures assistive screen readers announce input purpose to visually impaired users
(B) It styles labels in blue automatically
(C) It prevents duplicate entries
(D) It enables server database caching

85. In HTML5, which tag is used to specify multiple alternative image sources based on screen resolution and media queries for responsive images?

(A) <picture>
(B) <img>
(C) <source-img>
(D) <responsive>

86. In the <img> tag, how does the 'srcset' attribute combined with the 'sizes' attribute optimize mobile bandwidth?

(A) It allows browsers to choose and download the most appropriately sized image asset based on the device's display pixel density and viewport width
(B) It compresses images into WebP on the fly
(C) It converts SVG into Canvas raster
(D) It downloads all images simultaneously

87. What is the security hazard of opening external links with 'target="_blank"' without including 'rel="noopener noreferrer"'?

(A) The opened page can access and redirect the origin page via window.opener, enabling Tabnabbing phishing attacks
(B) The origin page runs out of memory
(C) The browser crashes instantly
(D) Cookies are exposed in plain text

88. In HTML5 Canvas, how do you obtain the 2D rendering context required to draw lines, rectangles, and text?

(A) const ctx = canvas.getContext("2d");
(B) const ctx = canvas.getRenderContext();
(C) const ctx = canvas.createContext2D();
(D) const ctx = new CanvasRenderingContext2D(canvas);

89. What happens when you draw graphics on an HTML5 canvas whose CSS width and height differ from its internal attribute width and height?

(A) The canvas raster graphics will appear stretched, blurred, or distorted because CSS scales the raster buffer rather than expanding canvas resolution
(B) The browser throws an exception
(C) The canvas reverts to 300x150
(D) The canvas becomes completely transparent

90. Which HTML5 element represents content that has been deleted or strikethrough from a document, paired with <ins> for additions?

(A) <del>
(B) <strike>
(C) <s>
(D) <remove>

91. Which element represents text highlighted or marked for reference purposes, rendering with a yellow background by default?

(A) <mark>
(B) <highlight>
(C) <yellow>
(D) <b>

92. Which HTML5 attribute allows any webpage text content to be edited directly by the user in real time inside the browser?

(A) contenteditable="true"
(B) editable="true"
(C) user-modify="read-write"
(D) allow-edit="true"

93. What does the HTML5 'draggable="true"' attribute enable on an element?

(A) Enables native Drag and Drop API events (dragstart, dragover, drop) to be fired on the element
(B) Enables CSS transform animations
(C) Allows resizing via mouse dragging
(D) Scrolls element into view

94. Which meta tag setting is mandatory in responsive web design to ensure proper scaling across mobile viewports?

(A) <meta name="viewport" content="width=device-width, initial-scale=1.0">
(B) <meta name="responsive" content="true">
(C) <meta name="screen" content="mobile">
(D) <meta name="scale" content="1">

95. What is the primary difference between HTML5 Web Workers and standard JavaScript execution?

(A) Web Workers run script tasks in background worker threads, preventing heavy CPU computations from blocking the main UI rendering thread
(B) Web Workers have full DOM access
(C) Web Workers run on the web server
(D) Web Workers replace CSS animations

96. What is the restriction on HTML5 Web Workers regarding DOM access?

(A) Web Workers do not have access to the 'document', 'window', or DOM elements; they communicate with the main thread strictly via postMessage()
(B) They cannot use Math functions
(C) They cannot execute loops
(D) They cannot make HTTP requests

97. What is the role of an HTML5 Application Cache manifest (cache.manifest) historically, and what modern technology replaced it?

(A) It cached web assets for offline usage; it has been entirely deprecated and replaced by Service Workers and Cache API
(B) It cached SQL databases; replaced by IndexedDB
(C) It compressed CSS; replaced by Sass
(D) It encrypted cookies; replaced by JWT

98. In the HTML5 <form> element, what does the 'novalidate' boolean attribute do?

(A) It bypasses all native browser client-side constraint validation upon form submission (used when custom JS validation handles everything)
(B) It disables server validation
(C) It encrypts form values with AES
(D) It submits form without refreshing

99. Which HTML5 form attribute allows an <input> or <button> inside a form to submit to a different URL than the form's 'action' attribute?

(A) formaction
(B) formmethod
(C) formenctype
(D) formtarget

100. What is the difference between synchronous script loading (<script src="...">), 'async', and 'defer' attributes in HTML5?

(A) 'async' downloads script in parallel and executes immediately upon download, interrupting HTML parsing; 'defer' downloads in parallel but executes in order only AFTER HTML parsing finishes
(B) 'async' blocks parsing; 'defer' does not
(C) 'defer' executes before DOM loads
(D) 'async' only works for inline scripts

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Explain the foundational structure of an HTML5 document, detailing each core tag from <!DOCTYPE html> to closing </html>.

Q2. Compare HTML4 block layout elements (,) with HTML5 semantic elements (<header>, <nav>, <main>, <article>, <section>, <aside>, <footer>).

Q3. Explain the structure and implementation of HTML Tables, detailing <thead>, <tbody>, <tfoot>, colspan, rowspan, and accessibility captioning.

Q4. Describe the HTML Form architecture, explaining the <form> element attributes: action, method, enctype, and novalidate.

Q5. Categorize and explain the complete set of HTML5 Form Input Types with appropriate code examples and use cases.

Q6. Explain HTML5 form validation attributes: required, pattern, min, max, minlength, maxlength, and custom validation via JavaScript.

Q7. Explain native multimedia integration in HTML5 using <audio> and <video> elements, discussing formats, codecs, and fallback strategies.

Q8. Compare HTML5 <canvas> and Scalable Vector Graphics (<svg>), detailing their architectures, advantages, and use cases.

Q9. Explain HTML5 Web Storage APIs (localStorage vs sessionStorage) and contrast them with traditional HTTP Cookies.

Q10. Describe Responsive Web Design foundational HTML principles, including viewport configuration and the <picture> element.

Q11. Explain Web Accessibility (A11y) standards, the WAI-ARIA framework, and HTML practices for screen reader compatibility.

Q12. Explain the HTML5 Geolocation API, detailing permission flows, methods, and position data properties.

Q13. Explain HTML5 Drag and Drop API architecture, detailing events, dataTransfer object, and drag workflows.

Q14. Describe HTML5 Web Workers, explaining why they are needed, how they communicate, and their technical limitations.

Q15. Explain the differences between asynchronous script loading mechanisms: default <script>, async, and defer.

Q16. Explain the <details> and <summary> disclosure widgets in HTML5, their attributes, and accessibility benefits.

Q17. Describe HTML5 Microdata and schema.org integration, explaining itemscope, itemtype, and itemprop for SEO.

Q18. Detail the security implications of iframe embedding, explaining clickjacking attacks and the 'sandbox' attribute.

Q19. Explain the HTML5 Content Security Policy (CSP) meta tag and its role in mitigating Cross-Site Scripting (XSS).

Q20. Describe the HTML5 History API and how it enables Single Page Applications (SPAs) without full page reloads.

Q21. Write a complete, valid HTML5 form markup implementing accessible input fields for a user registration portal with comprehensive validation.

Q22. Explain HTML5 Canvas rendering pipeline and write JavaScript code to draw a customized geometric composition (rectangle, circle, text).

Q23. Explain the technical role of Web Workers in computational tasks and write a complete main-thread and worker-thread script implementation.

Q24. Detail the differences between HTML elements , <section>, and <article>, formulating clear guidelines for when to choose each.

Q25. Explain the HTML5 Drag and Drop event flow and implement a drag-and-drop file upload zone using native JavaScript.

Q26. Discuss HTML5 internationalization (i18n) and localization features: the lang attribute, dir attribute, <bdi>, and <ruby> tags.

Q27. Explain the structure and operation of HTML5 Server-Sent Events (SSE) using the <EventSource> API and contrast with WebSockets.

Q28. Detail how HTML5 custom data attributes (data-*) function and how they are manipulated via CSS and the dataset API in JavaScript.

Q29. Explain the importance of HTML Head meta tags for Social Media optimization using Open Graph (OG) and Twitter Cards.

Q30. Analyze common HTML parsing pitfalls, Quirks Mode vs Standards Mode, and DOM tree construction rules.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[A]	**Q21:**	[A]	**Q41:**	[A]	**Q61:**	[A]	**Q81:**	[A]
Q2:	[A]	**Q22:**	[A]	**Q42:**	[A]	**Q62:**	[A]	**Q82:**	[A]
Q3:	[A]	**Q23:**	[A]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[A]	**Q24:**	[A]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[A]
Q5:	[A]	**Q25:**	[A]	**Q45:**	[A]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[A]	**Q26:**	[A]	**Q46:**	[A]	**Q66:**	[A]	**Q86:**	[A]
Q7:	[A]	**Q27:**	[A]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[A]	**Q28:**	[A]	**Q48:**	[A]	**Q68:**	[A]	**Q88:**	[A]
Q9:	[A]	**Q29:**	[A]	**Q49:**	[A]	**Q69:**	[A]	**Q89:**	[A]
Q10:	[A]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[A]	**Q31:**	[A]	**Q51:**	[A]	**Q71:**	[A]	**Q91:**	[A]
Q12:	[A]	**Q32:**	[A]	**Q52:**	[A]	**Q72:**	[A]	**Q92:**	[A]
Q13:	[A]	**Q33:**	[A]	**Q53:**	[A]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[A]	**Q34:**	[A]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[A]	**Q35:**	[A]	**Q55:**	[A]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[A]	**Q36:**	[A]	**Q56:**	[A]	**Q76:**	[A]	**Q96:**	[A]
Q17:	[A]	**Q37:**	[A]	**Q57:**	[A]	**Q77:**	[A]	**Q97:**	[A]
Q18:	[A]	**Q38:**	[A]	**Q58:**	[A]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[A]	**Q39:**	[A]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[A]	**Q40:**	[A]	**Q60:**	[A]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

1. Text & Identity Types:
- type="text": Standard single-line alphanumeric string entry.
- type="password": Obfuscates entered characters for credential protection.
- type="email": Validates against standard RFC email syntax and displays '@' on mobile keyboards.
- type="tel": Numeric keyboard for phone numbers.
- type="url": Validates complete URI schemes (e.g., https://).

2. Numeric & Range Types:
- type="number": Restricts input to numbers, supporting 'min', 'max', and 'step' attributes.
- type="range": Renders a continuous horizontal slider control for adjusting numeric levels.

3. Temporal & Visual Types:
- type="date", "time", "datetime-local": Activates native operating system calendar and clock pickers.
- type="color": Renders a native color swatch picker returning a 6-digit hex code (#RRGGBB).

4. Selection & File Types:
- type="checkbox", "radio": Binary toggles and mutually exclusive option sets.
- type="file": Triggers OS file selection dialogs, supporting 'multiple' and 'accept' MIME filters.

```
<source src="movie.mp4" type="video/mp4; codecs='avc1.42E01E, mp4a.40.2'">
<source src="movie.webm" type="video/webm; codecs='vp8, vorbis'">
<track src="subs.vtt" kind="subtitles" srclang="en" label="English">
Your browser does not support HTML5 video.
</video>
```

Attributes & Best Practices:

1. controls: Renders native playback, volume, and fullscreen controls.
2. autoplay, loop, muted: Automates playback; modern browsers strictly mandate 'muted' for autoplay to trigger.
3. poster: Displays an image thumbnail while video buffers.
4. <source> fallback: Browsers parse top-down and load the first compatible codec (MP4/H.264 or WebM/VP9).
5. <track>: Injects WebVTT (Web Video Text Tracks) subtitles and closed captions for universal accessibility.

>>Solution Question Bank: HTML5 WebStorage API: Local vs Session Storage, details, architecture, and HTTP Cookies cases.

cleared instantly when the tab or window is closed.

2. Contrast with Traditional HTTP Cookies:
- Size Limit: Cookies are restricted to 4KB per cookie; Web Storage allows 5-10MB.
- Network Overhead: Cookies are automatically appended to the HTTP header of EVERY single request made to that domain, creating wasted bandwidth on static asset fetches. Web Storage data is purely client-side and never sent across the network.
- Security: Cookies with HttpOnly flag prevent XSS theft. Web Storage is fully accessible to JavaScript, making it susceptible to XSS token theft.

- States & Properties (aria-expanded="true", aria-hidden="true", aria-required="true"): Dynamically communicates widget states.
- Live Regions (aria-live="polite"): Announces dynamic asynchronous text updates (e.g., chat messages or error alerts) without shifting focus.

>>Solution Question Bank & Solutions. Limitations.

HTML5 introduced native disclosure widgets that allow users to show or hide supplementary information without writing JavaScript accordion accordions:

Syntax:
```html
<details open>
<summary>Frequently Asked Questions (FAQ)</summary>
<p>Here are detailed answers to common user questions...</p>
</details>
```

Features & Operational Behavior:
1. <summary>: Serves as the interactive clickable toggle header, displaying a rotating disclosure triangle icon automatically.
2. 'open' Attribute: A boolean attribute indicating whether details are expanded or collapsed. Adding 'open' expands the widget by default.
3. Accessibility: Browsers natively expose the widget's expanded/collapsed state to screen readers via accessibility APIs, complete with built-in keyboard Space/Enter toggling.

>>Solution to Question 15: Describe the HTML5 modifications of the embedding element <object> and the introduction of SEO.

Applies strict security restrictions to embedded iframe content by default:
- `<iframe src="external.html" sandbox>`: Disables JavaScript execution, form submission, popups, cookie access, and cross-origin navigation.
- Granular Permissions: Developers can relax specific constraints safely: `sandbox="allow-scripts allow-forms allow-same-origin"`.

```
  <label for="username">Username (min 4 characters):</label>
  <input type="text" id="username" name="username" required minlength="4" maxlength="20"
  placeholder="e.g. john_doe">

  <label for="email">Email Address:</label>
  <input type="email" id="email" name="email" required placeholder="name@domain.com">

  <label for="pwd">Password (at least 8 chars, 1 number):</label>
  <input type="password" id="pwd" name="password" required pattern="^(?=.*[0-9]).{8,}$">

  <label for="dob">Date of Birth:</label>
  <input type="date" id="dob" name="dob" max="2008-01-01" required>

  <button type="submit">Create Account</button>
  </fieldset>
  </form>
  ```
```

---
```

Web Workers offload heavy computations to background threads. Below is a complete prime-number calculation example:

Main Script (main.js):

```javascript
const worker = new Worker('primeWorker.js');
worker.postMessage({ maxLimit: 1000000 });

worker.onmessage = function(event) {
document.getElementById('result').innerText = `Total Primes: ${event.data}`;
};
```

Worker Script (primeWorker.js):

```javascript
onmessage = function(e) {
const max = e.data.maxLimit;
let count = 0;
for (let i = 2; i <= max; i++) {
let isPrime = true;
for (let j = 2; j * j <= i; j++) {
if (i % j === 0) { isPrime = false; break; }
}
if (isPrime) count++;
}
postMessage(count);
};
```

>>Solution Question 25 The difference between DOM event flow and event propagation is that DOM event flow describes the overall process of how events travel through the document, while event propagation specifically refers to the bubbling and capturing phases within that flow. Here's a detailed explanation to help you choose each.

```javascript
const dropZone = document.getElementById('dropZone');

// Prevent browser default behavior of opening file as a page
['dragenter', 'dragover'].forEach(eventName => {
dropZone.addEventListener(eventName, (e) => {
e.preventDefault();
dropZone.classList.add('highlight');
});
});

dropZone.addEventListener('dragleave', () => dropZone.classList.remove('highlight'));

dropZone.addEventListener('drop', (e) => {
e.preventDefault();
dropZone.classList.remove('highlight');
const files = e.dataTransfer.files;
if (files.length > 0) {
uploadFiles(files); // Dispatch to backend via FormData
}
});
```

```

Contrast with WebSockets:
- Directionality: SSE is strictly Unidirectional (Server -> Client); WebSockets are Bidirectional full-duplex.
- Protocol: SSE runs over standard HTTP/1.1 or HTTP/2 without protocol upgrades, passing through enterprise proxies and firewalls effortlessly; WebSockets require protocol upgrade handshake.
- Reconnection: SSE provides native automatic reconnection and event ID tracking out of the box.

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```

```
<meta name="twitter:site" content="@EngineeringDev">
<meta name="twitter:creator" content="@Author">
```

Providing high-resolution images (1200x630 px) with accurate descriptions dramatically amplifies social click-through rates (CTR).

Solution to Question 30:

>> Solution to Question 30: Analyze common HTML parsing pitfalls, Quirks Mode vs Standards Mode, and DOM tree construction rules.